

Report
Of
C++ Programming Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: 25SCS1003000803

Name of Student: RAJKUMAR

Batch: 2025-29

CANDIDATE'S DECLARATION

I, RAJKUMAR, do hereby declare that the internship report titled “C++ Programming Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name: RAJKUMAR

Roll No.: 25SCS1003000803

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Thiranex (Skill Development & Future Tech) for providing me the opportunity and a structured platform to learn and gain hands-on exposure in the domain of C++ Programming. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: RAJKUMAR

Roll No.: 25SCS1003000803

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3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter



Date: 02 August 2026

TO,
Rajkumar
Intern ID: THX-AUG0226-220

Subject: Offer Letter for Internship in C++ Programming

Dear **Rajkumar**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - C++ Programming
Commencement Date	02 Aug 2026
Completion Date	01 Sept 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the C++ Programming Virtual Internship undertaken at Thiranex, a Skill Development & Future Tech organisation, from 02 August 2026 to 01 September 2026. The internship was carried out remotely on a project basis as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of the internship was to build a strong practical foundation in the C++ programming language – covering core syntax, control structures, functions, object-oriented programming, pointers and memory management, the Standard Template Library (STL), and file handling – and to apply these concepts by designing and developing a console-based mini project. The internship combined structured self-paced learning material with periodic mentor review and a project-based evaluation, culminating in the award of a Certificate of Achievement (Intern ID: THX-AUG0226-220) by Thiranex.

4.2 Organization Profile

Thiranex is a skill-development and future-technology focused organisation that designs and delivers industry-oriented virtual internship programs for engineering and technology students. Operating under the theme “Skill Development & Future Tech” and registered as an MSME entity, Thiranex offers remote, project-based internships across domains such as programming, software development, and emerging technologies. The organisation pairs self-paced learning modules with periodic progress reviews and industry mentorship, and issues a verifiable digital Certificate of Achievement (with a QR-based verification ID) on successful completion. The internship program is led by Mr. Hariharan M, Founder & CEO of Thiranex.

4.3 Problem Statement

Although C++ is taught as a core subject in most undergraduate Computer Science curricula, students typically gain only theoretical exposure to concepts such as object-oriented design, dynamic memory management, and the Standard Template Library, with limited opportunity to apply them to a complete, working application. The problem

addressed by this internship was to bridge that gap by practically learning the C++ language from first principles through to intermediate concepts, and by applying that knowledge to design, code, test, and document a functional console-based application.

4.4 Project Objectives

- To understand and apply the fundamentals of C++ – syntax, data types, operators, control structures, and functions.
- To learn and implement object-oriented programming concepts such as classes, objects, constructors, inheritance, and polymorphism.
- To understand pointers, references, and dynamic memory allocation in C++.
- To gain hands-on experience with the Standard Template Library (STL), including vectors, maps, and iterators.
- To implement file handling for persistent storage of program data.
- To design, develop, and test a console-based mini project applying the above concepts.
- To follow good programming practices, including modular code design and basic debugging techniques.
- To successfully complete all project milestones and earn the internship completion certificate.

4.5 Scope of the Project

The scope of this internship spans the core-to-intermediate C++ programming workflow – starting with language fundamentals and control flow, moving through object-oriented programming and pointer-based memory management, and extending into STL usage and file-based data persistence. These concepts were consolidated through the design and development of a console-based Student Record Management System capable of adding, viewing, searching, updating, and storing student records using file handling. The internship was limited to a virtual, self-paced learning and project environment provided by Thiranex, and did not involve deployment on any live production or organisational system.

4.6 Technologies and Tools Used

- Programming Language: C++ (C++11/14 standard)
- Compiler/Toolchain: GNU G++ Compiler (GCC)
- IDE/Editor: Visual Studio Code / Code::Blocks
- Core Libraries: Standard Template Library (STL) – vector, map, string, algorithm
- Concepts Applied: Object-Oriented Programming (OOP), Pointers and Dynamic Memory Management, File Handling (fstream)
- Version Control: Git (for tracking project progress)
- Learning Platform: Thiranex self-paced virtual internship modules with mentor review

4.7 System Architecture

The console-based mini project developed during the internship follows a simple layered architecture, summarised below:

- Presentation Layer: A menu-driven console interface that accepts user input and displays output/results.
- Business Logic Layer: C++ classes and functions implementing the core operations – adding, searching, updating, and deleting records – using object-oriented design principles.
- Data Layer: File handling (using fstream) to persist records to, and retrieve records from, a local data file so that information is retained between program runs.
- Utility Layer: STL containers and algorithms (vector, map, sorting/searching routines) used to manage records efficiently in memory before they are written to file.

4.8 Methodology

The internship followed a structured, module-wise methodology combining conceptual learning, hands-on coding exercises, and a final mini project, as summarised in the table below.

Week	Module	Description
1	C++ Fundamentals	Syntax, data types, operators, control structures (if/else, loops), and functions; setting up the compiler/IDE and writing first programs.
2	Object-Oriented Programming	Classes and objects, constructors/destructors, encapsulation, inheritance, and polymorphism through hands-on exercises.
3	Pointers, Memory & STL	Pointers and references, dynamic memory allocation (new/delete), and the Standard Template Library – vectors, maps, and iterators.
4	File Handling & Mini Project	File handling with fstream for persistent storage; design, coding, and testing of the Student Record Management System mini project.
Final	Project Review & Assessment	Mentor review of the completed project and final assessment leading to the award of the Certificate of Achievement.

Note: The internship was conducted in remote/project-based mode with periodic mentor review, and did not carry a stipend.

4.9 Expected Outcomes

- Ability to write, compile, and debug C++ programs using core language constructs.
- Practical understanding of object-oriented programming – classes, inheritance, and polymorphism.
- Working knowledge of pointers, dynamic memory management, and common pitfalls to avoid.
- Ability to use STL containers and algorithms to write efficient, reusable code.
- Hands-on experience with file handling for persistent data storage.
- A completed, working console-based mini project (Student Record Management System) demonstrating the above skills.
- Successful completion of the internship with a verifiable Certificate of Achievement from Thiranex.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Certificate of Achievement issued by Thiranex, confirming successful completion of the C++ Programming Virtual Internship (Intern ID: THX-AUG0226-220) conducted from 02 August 2026 to 01 September 2026, are attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

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